

AKILAN M

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PROFESSIONAL SUMMARY

Data Scientist skilled in building **end-to-end machine learning pipelines**, **NLP systems**, and **time series forecasting models**. Experienced in developing, deploying, and monitoring ML applications using Python, Scikit-learn, FastAPI, Streamlit, and Hugging Face Spaces. Passionate about building scalable AI systems that solve real-world business problems.

TECHNICAL SKILLS

Python, SQL | Machine Learning: Regression, Classification, Decision Trees, Random Forest, SVM | **Deep Learning:** TensorFlow, Pytorch CNN, LSTM, GRU | **NLP, Time Series & LLM:** TF-IDF, Embeddings, Sentiment Analysis, ARIMA, Topic Modeling (Bertopic), Transformers, RAG | **MLOps & Deployment:** FastAPI, Streamlit, Render, Hugging Face Spaces, Git, GitHub | **Data Visualization:** Matplotlib, Seaborn, Plotly, Power BI

PROJECTS

Customer Churn Prediction MLOps System

Tech: Python, Scikit-learn, UMAP, HDBSCAN, LLM, FastAPI, Streamlit

GitHub: <https://github.com/akilanm71/customer-churn-mlops-system>

Live App: https://huggingface.co/spaces/akilanjai71/churn_prediction_labeling

Built an end-to-end ML pipeline for telecom churn prediction including preprocessing, feature engineering, model training, and evaluation.

Implemented UMAP dimensionality reduction and HDBSCAN clustering to identify hidden patterns in customer complaints.

Developed an LLM-based automated topic labeling system to organize complaints into interpretable segments.

Designed FastAPI-based model inference APIs and deployed services using cloud platforms.

Deployed an interactive application on Hugging Face Spaces for real-time churn prediction and feedback analysis.

Customer Churn Analytics Dashboard

Tech: Python, Pandas, Streamlit, Plotly

GitHub: https://github.com/akilanm71/customer_churn_dashboard

Live App: <https://customerchurndashboard-82dgvxx9eqz6grv5kgkfbu.streamlit.app>

Developed an **interactive analytics dashboard** to visualize telecom churn patterns.

Built dynamic charts analysing customer tenure, service usage, and churn distribution.

Integrated model outputs to highlight high-risk churn segments for business insights.

Electricity Production Forecasting | ARIMA, LSTM, GRU

GitHub: <https://github.com/akilanm71/Electricity-production-forecast>

Developed time series forecasting models to predict electricity production trends.

Implemented and compared **ARIMA, LSTM, and GRU architectures**.

Performed stationarity testing, data transformation, and model evaluation.

CERTIFICATIONS / TRAINING

Data Scientist (**NASSCOM**) | Python for Data Analysis & Visualization (**Udemy**) | Python Bootcamp (**Udemy**) | AWS Cloud Practitioner (**AWS**) | Certified Data Scientist – **DataMites**, Bangalore (July 2025 – Present)

WORK EXPERIENCE

Senior Executive – Pharmaceutical Operations | **Biocon Biologics** | Bangalore | 2023–2025

- Analyzed large-scale production and quality datasets using Excel to identify trends and process inefficiencies.
- Converted raw operational data into structured reports and summaries to support data-driven decision-making.
- Monitored process metrics and compliance indicators, ensuring data accuracy and validation across workflows.

EDUCATION

M-Tech - BioTechnology

National Institute of Technology Warangal, Telangana 2023 | **CGPA:** 7.28

B-Tech - BioTechnology

Kalasalingam Academy of Research and Education, TamilNadu 2015